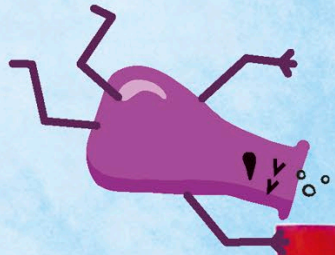


Take to the skies

In order to fly, you have to overcome the force of gravity pulling you towards the ground. Helicopters and aeroplanes use rotors, wings, and engines to soar through the air.



Drag

Drag, or air resistance, is the force that pulls the helicopter backwards. Drag increases as the helicopter moves faster.

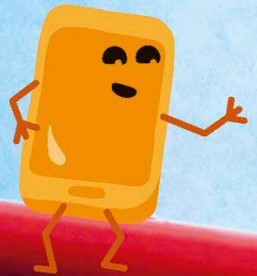
Forces of flight

There are four main forces working on a helicopter as it flies. Drag tries to slow it down, gravity tries to bring it back to Earth, lift raises it upwards, and thrust propels it forwards.

Tail rotor

Without a tail rotor the helicopter would turn in circles!

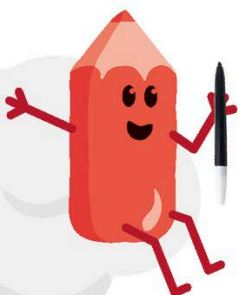
In a hovering helicopter, all the forces balance exactly.



Helicopters can fly backwards!

Dreams of flying

The Italian artist and inventor Leonardo da Vinci was fascinated with flying. He studied birds and drew many imaginary flying devices, such as a wing-flapping machine for humans.



Da Vinci's wing design